

DATADOSE PROJECT DOCUMENTATION



AI-powered polypharmacy safety platform



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1. Project Overview

DataDose is an enterprise-grade AI-powered Clinical Decision Intelligence platform that delivers polypharmacy safety at the point of care. It combines a Neo4j drug-interaction knowledge graph, real-time streaming, and generative AI, backed by a production-quality data engineering pipeline running from raw pharmacy records to a governed Snowflake analytics warehouse, containerized end-to-end and orchestrated by Apache Airflow.

The platform operates across three integrated layers: a Live Clinical Application (Next.js 15 + FastAPI + Neo4j AuraDB) providing real-time drug interaction scanning, visual prescription mapping, and an AI clinical assistant across five role-based dashboards, running as Docker containers; a Data Engineering Platform running a batch and streaming pipeline from raw CSV through pandas cleaning, Aiven Kafka, Databricks PySpark enrichment, a Snowflake star schema, to Power BI reporting; and

an Orchestration Layer where Apache Airflow, with its own Postgres metadata database, triggers, schedules, and monitors every pipeline stage.

Both application and pipeline layers model the same pharmaceutical knowledge base — Drug, Disease, and Symptom nodes connected by INTERACTS_WITH, TREATS, and CAUSES_REACTION relationships — keeping clinical intelligence and analytics conceptually unified.

2. Problem Statement

Polypharmacy — patients taking five or more concurrent medications — affects over 40% of elderly patients and is a leading cause of preventable adverse drug events. Clinicians face specific gaps:

- **No Real-Time DDI Checking:** Clinicians lack real-time, context-aware tools to check drug interactions across a patient's full medication list during a clinical encounter.
- **Fragmented Analytics:** Raw pharmacy data is fragmented, unvalidated, and siloed from analytics infrastructure, making population-level risk analysis slow and unreliable.
- **Handwritten Prescription Gaps:** Handwritten or printed prescriptions are not easily digitized into safety-checking workflows.
- **Limited Multilingual Support:** Existing tools rarely trace a reported symptom back to a likely causative drug across both English and Arabic terminology.
- These gaps lead to preventable adverse drug events, delayed interventions, and poor visibility into population-level prescribing risk.

3. Solution Overview

DataDose addresses these gaps with a unified platform that:

- **Scans Polypharmacy in Real Time:** An N-degree scanner checks all pairwise INTERACTS_WITH relationships among a patient's drug list in milliseconds via Neo4j Cypher traversal.
- **Grounds a GraphRAG Chatbot:** Routes clinician queries through Neo4j first for deterministic retrieval, then falls back to LLaMA-3 synthesis with a hallucination guard.
- **Suggests Safe Alternatives:** Cypher traversal finds substitute drugs that treat the same disease without triggering known interactions, with a Groq LLM fallback.
- **Extracts Drugs via OCR:** A vision LLM (LLaMA 3.2 11B) extracts drug names from uploaded prescription images.
- **On the analytics side,** DataDose runs a five-stage validation pipeline (pandas → OpenFDA → Groq verification → Kafka → Databricks) that lands clean, risk-scored prescription events in a governed Snowflake star schema, ready for executive Power BI dashboards, with every stage triggerable from Airflow.

4. Project Objectives

The primary objectives of the DataDose project include:

- **Deliver real-time N-degree DDI scanning:** Sub-second pairwise polypharmacy checks via Neo4j Cypher traversal.
- **Build a hybrid GraphRAG chatbot:** Grounding LLM responses in verified pharmaceutical knowledge from Neo4j.
- **Provide a Cypher-powered safe-alternative finder:** With a Groq LLM fallback when the graph returns zero results.
- **Extract drug names from prescription images:** Via a vision LLM OCR pipeline.
- **Trace symptoms back to causative drugs:** With English + Arabic synonym expansion.
- **Visualize the patient drug graph:** In an interactive React Flow canvas.
- **Build a five-stage cleaning and enrichment pipeline:** pandas → OpenFDA → Groq → Kafka → Databricks.
- **Land risk-scored events in Snowflake:** In a governed star schema for BI reporting.
- **Orchestrate the pipeline through Airflow:** Cleaning, streaming, and Databricks triggers via DAGs.
- **Containerize the live application with Docker:** For consistent local deployment of the frontend and backend.

5. System Architecture

DataDose's architecture spans three coordinated layers, with only the frontend, backend, and Airflow stack running locally in Docker; Neo4j, Snowflake, Aiven Kafka, and Databricks stay cloud-hosted:

- **Application Layer:** Next.js 15 (TypeScript, App Router) frontend and FastAPI (Python) backend, communicating over REST, both running as Docker containers.
- **AI & Knowledge Graph:** Neo4j AuraDB stores the Drug / Disease / Symptom graph (3,656+ nodes); Groq-hosted LLaMA-3 models power chat, OCR, and GraphRAG synthesis.
- **Data Engineering & Analytics:** Aiven-managed Apache Kafka on Aiven streams prescription events; Databricks PySpark enriches them against Neo4j and applies risk scoring; Snowflake hosts the star-schema warehouse; Power BI serves executive reporting.
- **Orchestration:** A local Apache Airflow instance (webserver + scheduler + its own Postgres metadata DB) triggers and monitors the cleaning, Kafka, and Databricks stages via DAGs.
- Neo4j AuraDB, Snowflake, Aiven Kafka, and Databricks are managed cloud services; only the frontend, backend, and Airflow stack run locally in Docker.

6. Neo4j Knowledge Graph

The Neo4j AuraDB knowledge graph is the clinical reasoning core of DataDose, holding 3,656+ nodes across three types:

- **Drug:** name, active ingredient, trade names.
- **Disease / Condition:** name, ICD code.
- **Symptom:** name, Arabic name for multilingual synonym matching.

These are connected by three relationship types: INTERACTS_WITH (Drug → Drug), carrying severity and mechanism type; TREATS (Drug → Disease), carrying efficacy level; and CAUSES_REACTION (Drug → Symptom), carrying frequency and severity. This structure powers sub-second pairwise polypharmacy checks, safe-alternative discovery, and the GraphRAG chatbot's grounded retrieval step.

Example Use Case: Given a patient's drug list, traverse all pairwise INTERACTS_WITH relationships to flag high-severity combinations in milliseconds.

7. Data Pipeline

DataDose runs a five-stage pipeline, triggerable end-to-end from Airflow:

1. **Stage 1 — Raw Ingestion:** raw pharmacy CSV records (drug names, active ingredients, diseases, dosages) become the source of truth for both the graph and the warehouse.
2. **Stage 2 — Pandas Cleaning (notebooks 01–05):** column standardization and deduplication; active-ingredient fuzzy matching and regex normalization; OpenFDA + Groq LLM ingredient verification; trade-name cleaning; CSV merge into output_FINAL.csv.
3. **Stage 3 — Kafka Streaming:** the producer reads output_FINAL.csv and emits synthetic prescription events as JSON to Aiven Kafka (SASL_SCRAM-SHA-256 + TLS), topic DataDose.in.
4. **Stage 4 — Databricks PySpark Enrichment:** Structured Streaming consumes Kafka events, queries Neo4j for each drug's interactions, applies a risk-scoring function, and writes to Snowflake via the Spark connector. Stage 5 — Snowflake Star Schema: a four-layer warehouse — **STAGING; DIMENSIONS (DIM_DRUG, DIM_INGREDIENT, DIM_PHARMACY, DIM_PATIENT, DIM_DATE, DIM_DRUG_INGREDIENT), FACTS (FACT_PRESCRIPTION_TRANSACTION)**, and ANALYTICS views.

8. Real-time Data Streaming

DataDose uses **Apache Kafka on Aiven** as its real-time event backbone. The producer reads verified drug records from output_FINAL.csv and publishes JSON prescription events to the DataDose.in topic over SASL_SSL.

Databricks Structured Streaming consumes these events, queries Neo4j for INTERACTS_WITH relationships and severity scores for each drug in the prescription, applies risk scoring, and writes the enriched rows into Snowflake — moving them through the staging, dimension, fact, and analytics layers.

9. Analytics Warehouse

DataDose's analytics warehouse is a governed **Snowflake star schema**, built and populated as the final stage of the streaming pipeline: STAGING holds raw inbound prescription events; DIMENSIONS

covers DIM_DRUG , DIM_INGREDIENT, DIM_PHARMACY, DIM_PATIENT, DIM_DATE, and the DIM_DRUG_INGREDIENT bridge table; FACTS covers FACT_PRESCRIPTION_TRANSACTION; and ANALYTICS holds aggregated risk and prescription views feeding Power BI.

The schema DDL lives in Snowflake, with connection setup documented in Snowflake/README.md.

10. Airflow Orchestration

A **local Apache Airflow instance** (webserver + scheduler, backed by its own PostgreSQL metadata database, both running in Docker) is the control plane for the DataDose data pipeline and other local tooling. Suggested DAGs:

cleaning pipeline runs the Cleaning Code/01–05 notebooks in order, producing output_FINAL.csv. Kafka producer starts/stops producer_simulator.py runs or triggers a fixed-size batch. databricks_trigger kicks off the Databricks PySpark job via the Databricks Jobs API, scheduled or downstream of kafka_producer. local_tooling wraps ad-hoc maintenance scripts such as dataset refresh and Snowflake verification queries.

The Airflow webserver UI at localhost:8080 gives full visibility into DAG runs, logs, and retries across the whole pipeline.

11. AI Clinical Features

DataDose integrates six AI-powered clinical features, all backed by Groq-hosted LLaMA-3 models:

- **Hybrid GraphRAG Chatbot:** routes a clinician's question through entity extraction, Neo4j graph retrieval, then LLaMA-3 synthesis with a hallucination guard.
- **Prescription OCR Scanner:** Groq's llama-3.2-11b-vision-preview vision model extracts drug names from handwritten or printed prescription images, feeding them directly into the DDI scanner.
- **Smart Safe Alternatives:** multi-constraint Cypher traversal finds substitute drugs that treat the same disease without triggering known interactions or symptoms, with a Groq LLM fallback when the graph returns zero results.
- **Reverse Symptom Tracer:** a wide-net tracer with an English + Arabic SYMPTOM_SYNONYMS dictionary, tracing a reported symptom back to the most likely causative drug among the patient's current medications.
- **Real-Time DDI Scanner:** N-degree polypharmacy scanner checking all pairwise INTERACTS_WITH relationships, sorted by severity in milliseconds via Cypher.
- **Visual Prescription Map:** a React Flow canvas rendering a patient's drug list as an interactive graph of DDI edges, treated diseases, and caused symptoms.

12. Technology Stack

The DataDose platform is built using a modern and robust technology stack:

- **Frontend:** Next.js 15, React 19, TypeScript, Tailwind CSS v4, Framer Motion
- **Data Visualization:** React Flow, Chart.js, Recharts
- **Auth & Data Layer:** NextAuth, Prisma ORM, PostgreSQL
- **Backend API:** FastAPI, Python 3, Uvicorn
- **AI:** Groq (LLaMA-3), LLaMA 3.2 11B Vision
- **Graph:** Neo4j AuraDB, Cypher
- **Streaming:** Apache Kafka, Aiven
- **Data Engineering:** Pandas, NumPy, Databricks, PySpark
- **Warehousing & BI:** Snowflake, Power BI
- **Containerization & Orchestration:** Docker, Docker Compose, Apache Airflow
- **Testing:** Playwright

This stack provides a balance of performance, scalability, and developer productivity.

13. Power BI Reporting

Microsoft Power BI is utilized to create interactive and insightful reports and dashboards for clinicians and administrators. These visualizations provide a user-friendly interface to explore the data processed by DataDose.

Report Pages (DrugData.pbix, 7 pages):

Home: static cover page.

Overview: population-level prescription summary across the cleaned drug dataset (12,105 records, 18 fields).

Clinical Insights: drug interaction patterns and trends.

Risk Analysis: risk-score distribution and safety alerts, built on the Risk_Score and Total_Safety_Burden DAX measures.

Power BI connects to Snowflake via the built-in connector (Import or DirectQuery), reading from the PHARMA_ANALYTICS_DB warehouse.

14. Docker Deployment

To ensure consistent, reproducible, and scalable deployments, DataDose leverages **Docker** containerization. Only the components that run locally are containerized: the Next.js frontend, the FastAPI backend, and the Airflow webserver/scheduler with its own Postgres metadata database. Neo4j AuraDB, Snowflake, Aiven Kafka, and Databricks remain managed cloud services and are not containerized.

Benefits of Docker Deployment:

- **Environment Consistency:** Eliminates 'it works on my machine' issues.
- **Scalability:** Easily scale services up or down by launching more containers.
- **Isolation:** Services run in isolated environments, preventing conflicts.
- **Simplified Management:** Streamlines installation, updates, and rollbacks.

docker-compose.yml wires the frontend and backend containers together on a shared network; a separate docker-compose.airflow.yml runs the Airflow stack alongside its Postgres metadata database.

15. Key Performance Indicators (KPIs)

The success of the DataDose platform is measured against several Key Performance Indicators (KPIs):

- **DDI Scan Latency:** Sub-second response time for pairwise Neo4j Cypher polypharmacy checks.
- **Pipeline Throughput:** Kafka producer rate (messages/sec) and Databricks micro-batch processing time from DataDose.in to Snowflake staging.
- **Data Quality:** Dead-letter rate in the Databricks pipeline (schema_parse_failed, enrichment_failed, snowflake_write_failed, kafka_out_write_failed) and Airflow data-quality check pass rate.
- **High-Risk Detection Rate:** Share of transactions flagged HIGH_RISK_PATIENT (PATIENT_RISK_SCORE $\geq$ 60) or Major/Moderate severity interactions.
- **Pipeline Reliability:** Successful Airflow DAG run rate at the 2-minute schedule, and Kafka producer uptime.
- **Warehouse Freshness:** Time from a Kafka event landing in STG_TRANSACTION to promotion into the dimensional model.

16. Conclusion

DataDose demonstrates how graph database technology, real-time streaming, and generative AI can be combined into a single Clinical Decision Intelligence platform — delivering sub-second polypharmacy safety checks at the point of care while feeding a governed Snowflake analytics warehouse for population-level insight. With a Neo4j knowledge graph of 3,656+ nodes, a five-stage Airflow-orchestrated data pipeline, six AI-powered clinical features, and a 4-page Power BI reporting layer, the project shows a complete path from raw pharmacy records to actionable clinical and executive intelligence — architected for containerized, reproducible deployment end to end.